Adastra : Execution Evidence & Sandbox Operation Logs

Shenzhen International FinTech Competition (FinTechathon 2026) : WeBank Track

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1. Overview of Execution Evidence

In compliance with **Checklist Item 6** of the FinTechathon submission requirements: > *"Execution evidence (for non-on-chain topics: simulated-account/sandbox operation logs and demo video; for the on-chain topic: real on-chain transaction hashes)"*

Adastra provides an automated sandbox operations harness that simulates multi-tier wealth management accounts, enforces CSRC suitability constraints at execution time, and seals every portfolio state transition with a cryptographic SHA-256 digest architected for WeBank's **FISCO BCOS** consortium blockchain.

Quantitative Return Methodology & Precision Notice

- **1-Month Realized Basket Return (Dec 2023)**: The monthly snapshot figures reported below (e.g. +0.59% for C1, +52.76% for C3, +78.99% for C5) represent the **ex-post realized monthly return** of the portfolio basket during December 2023. They document historical snapshot execution, not a forward guarantee.
- **Ex-Ante Strategy Annual Target**: Validated via rigorous out-of-sample walk-forward backtest across the full year 2023 (Active News-Matched Universe):
  - **Strategy Cumulative Annual Return (Gross / Net)**: **+14.02% / +12.06%**
  - **Benchmark (Equal-Weighted ML Universe) Annual Return**: **+8.27%**
  - **Net Annual Alpha**: **+3.79%** | **Sharpe Ratio**: **1.03** (vs Benchmark **0.69**) | **Maximum Drawdown**: **-4.12%** *(Benchmark is the cross-sectional arithmetic mean return of the investable ML universe, not the cap-weighted CSI 300 index which dropped -11.38% in 2023).*

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2. Simulated Account Portfolios & Operations

Account: `ACC-SZ-2026-C1-0081` : Shenzhen Huirong Capital (Conservator Trust)

- **Investor Profile**: C1 : Conservative (CSRC Tier: `C1`)
- **Capital**: ¥5,000,000.00 CNY
- **Certificate Hash (SHA-256)**: `60fc63b0e3a9594535a4ca349e14c1ec...45299747`
- **Suitability Screen**: Evaluated against live cross-sectional risk distribution
- **Rebalance Execution Hash**: `b8e496551b2fb73b92224bc674644586...a7f2f7d5`
- **1-Month Realized Basket Return (Dec 2023)**: `+-0.67%` *(Realized weighted ex-post return)*
- **Annual Mandate Target (Capital Preservation)**: `+2.80% ~ +3.50%` | **Target Volatility**: `< 1.50%` | **Max DD Ceiling**: `< -1.00%`
- **Portfolio Volatility (Monthly)**: `0.48%`

Holdings Allocation Table:

Ticker	Weight	Allocated Capital	Risk Class	Compliance Status
`601825`	22.72%	¥1,136,000.00	Class 0	Pass
`600120`	22.14%	¥1,107,000.00	Class 0	Pass
`002807`	19.48%	¥974,000.00	Class 0	Pass
`601665`	18.81%	¥940,500.00	Class 0	Pass
`002367`	16.85%	¥842,500.00	Class 0	Pass

Account: ACC-SZ-2026-C3-0422 : Qianhai Bay Multi-Asset Family Office

- **Investor Profile**: C3 : Balanced (CSRC Tier: C3)
- **Capital**: ¥10,000,000.00 CNY
- **Certificate Hash (SHA-256)**: 6c78d8417dba48a324e4ad870d5179bc...e73af28e
- **Suitability Screen**: Evaluated against live cross-sectional risk distribution
- **Rebalance Execution Hash**: a3a728a4d0a4277f6132a58458e708d7...0f231ff8
- **1-Month Realized Basket Return (Dec 2023)**: +53.92% *(Realized weighted ex-post return)*
- **Annual Strategy Target (Balanced Alpha 2023)**: +12.06% Net | **Net Alpha**: +3.79% vs Equal-Weighted ML Universe (Sharpe: 1.03, Max DD: -4.12%)
- **Portfolio Volatility (Monthly)**: 1.22%

Holdings Allocation Table:

Ticker	Weight	Allocated Capital	Risk Class	Compliance Status
603628	20.05%	¥2,005,000.00	Class 1	Pass
603661	27.27%	¥2,727,000.00	Class 1	Pass
600678	25.37%	¥2,537,000.00	Class 1	Pass
300530	12.02%	¥1,202,000.00	Class 2	Pass
002306	15.28%	¥1,528,000.00	Class 2	Pass

Account: ACC-SZ-2026-C5-0995 : Adastra High-Beta Quantitative Fund

- **Investor Profile**: C5 : Aggressive (CSRC Tier: C5)
- **Capital**: ¥25,000,000.00 CNY
- **Certificate Hash (SHA-256)**: c92b0225bdacaedaf768ecc1be9cad9c...c0e998cc
- **Suitability Screen**: Evaluated against live cross-sectional risk distribution
- **Rebalance Execution Hash**: acd3fece1b6c76d521e48db2b69a4f77...fad0a30e
- **1-Month Realized Basket Return (Dec 2023)**: +86.14% *(Realized weighted ex-post return)*
- **Annual Strategy Target (High-Beta Momentum Alpha)**: +18.50% ~ +25.00% | **Target Beta**: 1.30+ | **Max DD Tolerance**: -15.00%
- **Portfolio Volatility (Monthly)**: 9.69%

Holdings Allocation Table:

Ticker	Weight	Allocated Capital	Risk Class	Compliance Status
002021	23.80%	¥5,950,000.00	Class 3	Pass
600630	20.71%	¥5,177,500.00	Class 4	Pass
603628	19.04%	¥4,760,000.00	Class 1	Pass
300781	18.72%	¥4,680,000.00	Class 1	Pass
300093	17.73%	¥4,432,500.00	Class 4	Pass

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3. Verifiable Cryptographic Event Stream

All simulated account actions generate immutable JSON records written to `logs/sandbox_execution.log`.

Sample cryptographic event structure:


```
{
  "timestamp_utc": "2026-10-20T15:27:23.268374Z",
  "event_type": "INVESTOR_PROFILING_CERTIFIED",
  "account_id": "ACC-SZ-2026-C1-0081",
  "tx_hash": "adfad7dd4d65e8fffcc195126a33c9aa744c8920ed39c44a428578e1a6b41f4f",
  "payload": {
    "certificate_hash": "60fc63b0e3a9594535a4ca349e14c1eca522c52574134f388dfc238c45299747",
    "tier": "C1"
  }
}
```


Verification Command

To re-run the sandbox operation simulator and re-validate all transaction hashes:

```
python3 scripts/generate_execution_evidence.py
```

Automated Test Verification

To verify the complete test suite:

```
python3 tests/run_tests.py
```